Basic **2-input AND gate** is modeled in **Verilog** across all three techniques.

1. Gate-Level (Structural) Modeling

```
module and_gate_level (a,b,y);
    input a;
    input b;
    output y;
    and (y, a, b);
endmodule
```

2. Dataflow Modeling

```
module and_dataflow (a,b,y);
    input a;
    input b;
    output y;
    assign y = a &b;
endmodule
```

3. BehavioralModeling

```
module and_behavioural(a,b,y);
    input a;
    input b;
    output y;
     reg y ;
    always @(a or b)
 begin
        if (a = = 1'b1 && b = = 1'b1)
            y = 1'b1;
        else
            y = 1'b0;
    end
endmodule
```

Basic **2-input OR gate** is modeled in **Verilog** across all three techniques.

1. Gate-Level (Structural) Modeling

```
module or_gate_level (a,b,y);
   input a;
   input b;
   output y;
   or (y, a, b);
endmodule
```

2. Dataflow Modeling

```
module or_dataflow (a,b,y);
   input a;
   input b;
   output y;
   assign y = a |b;
endmodule
```

3. BehavioralModeling

```
module or_behavioural(a,b,y);
   input a;
   input b;
   output y;
    reg y;
  always @(a or b)
 begin
       if (a == 1'b0&& b == 1'b0)
          y = 1'b0;
       else
          y = 1'b1;
   end
endmodule
```

Basic **2-input XOR gate** is modeled in **Verilog** across all three techniques.

1. Gate-Level (Structural) Modeling

```
module xor_gate_level (a,b,y);
    input a;
    input b;
    output y;
    xor (y, a, b);
 endmodule
```

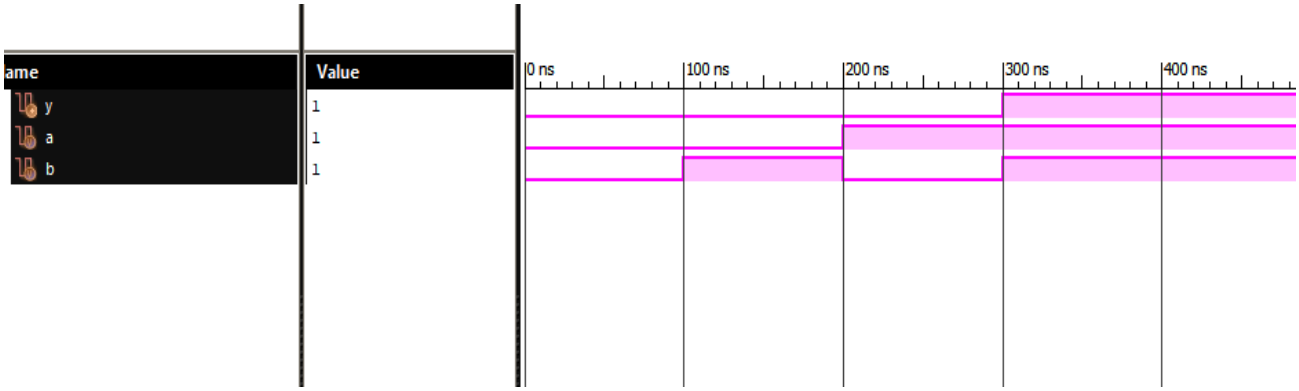
2. Dataflow Modeling

```
module xor_dataflow (a,b,y);
    input a;
    input b;
    output y;
    assign y = a ^b;
endmodule
```

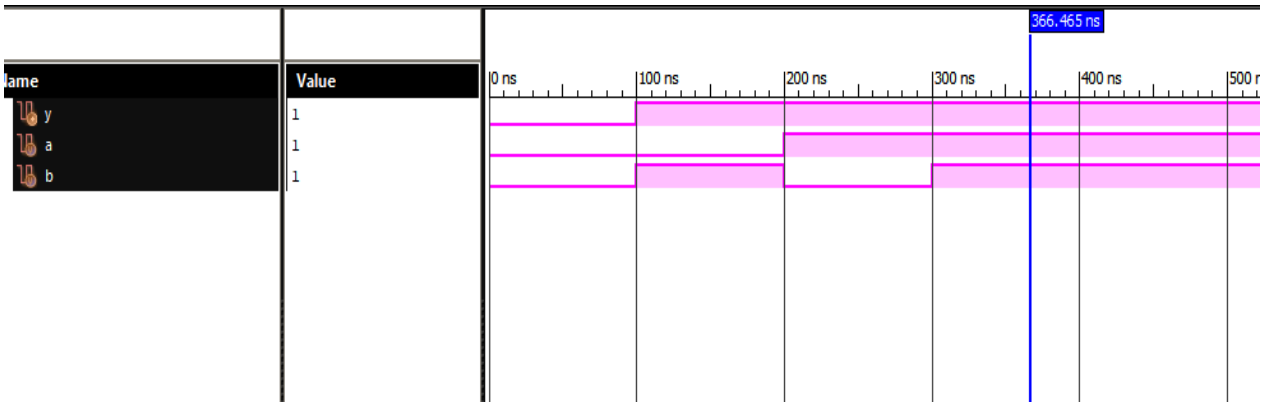
3. BehavioralModeling

```
module xor_behavioural(a,b,y);
    input a;
    input b;
    output y;
    output reg y;
    always @(a or b)
 begin
        if (a = = b)
            y = 1'b0;
        else
            y = 1'b1;
    end
endmodule
```

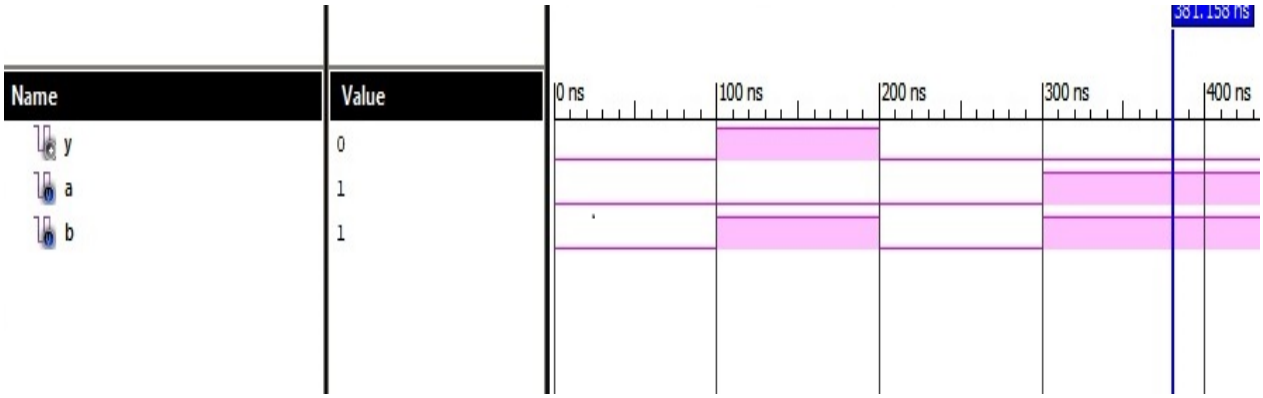
Output for AND gate



Output for OR gate



Output for XOR gate



VERILOG TEST FIXTURE

```
module and_gate_level;

	reg a;
	reg b;

	wire y;
	and_behavioural uut (
		.a(a),
		.b(b),
		.y(y)
	);

	initial begin
	// Initialize Inputs
		a = 0;
		b = 0;
		#100;

		a = 0;
		b = 1;
		#100;

		a = 1;
		b = 0;
		#100;

		a = 1;
		b = 1;
		#100;

	end
```